

5.1

$$2. \sum_{m=0}^{\infty} (m+1) m x^m$$

$$\frac{Z_{m+1}}{Z_m} = \frac{(m+2)(m+1)x^{m+1}}{(m+1)m x^m} = \frac{m+2}{m} \cdot x$$

$$L^{\infty} = \frac{1}{R} = 1$$

$$\underline{R=1}$$

$$6. (Hx)y' = 2y$$

$$\text{let } y = a_0 + a_1 x + a_2 x^2 + \dots$$

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + \dots$$

$$(Hx)(a_1 + 2a_2 x + 3a_3 x^2 + \dots) = 2(a_0 + a_1 x + a_2 x^2 + \dots)$$

$$x^0: a_1 = 2a_0$$

$$x^1: 2a_2 + a_1 = 2a_1$$

$$2a_2 = a_1 = 2a_0$$

$$a_2 = a_0$$

$$x^2: 3a_3 + 2a_2 = 2a_2$$

$$a_3 = 0$$

$$x^n: na_n + (n+1)a_{n+1} = 2a_n$$

$$a_{n+1} = \frac{2-n}{n+1} a_n$$

$$a_3 = 0$$

$$a_4 = \frac{-1}{4} a_3 = 0$$

⋮

$$\therefore y = a_0 + a_1 x + a_2 x^2 = a_0 (1 + x + x^2)$$

$$17. y'' + 4xy' + 2y = 0, y(0) = 1, y'(0) = 1 \quad x_1 = 0.25$$

$$let y = \sum_{n=0}^{\infty} a_n x^n$$

$$4xy' = \sum_{n=1}^{\infty} 4x n a_n x^{n-1} = \sum_{n=1}^{\infty} 4n a_n x^n$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$a_0 + a_1 x + a_2 x^2$$

$$a_1 + 2a_2 x$$

$$2a_2$$

$$\Rightarrow |3 \text{ 권 } 1 \text{ 권}| \quad a_0 = 1, a_1 = 1$$

$$x^0: 2a_0 + 2a_2 = 0 \rightarrow a_2 = -a_0 = -1$$

$$x^1: 2a_1 + 4 \cdot 1 \cdot a_1 + 3 \cdot 2a_3 = 0 \rightarrow a_3 = -a_1 = -1$$

⋮

$$x^5: 2a_5 + 4 \cdot 5a_5 + (5+2)(5+1)a_{5+2} = 0$$

$$\rightarrow a_{5+2} = -\frac{4 \cdot 5 + 2}{(5+2)(5+1)} a_5$$

$$a_4 = -\frac{4 \cdot 2 + 2}{4 \cdot 3} a_2 = -\frac{5}{6} a_2 = \frac{5}{6}$$

$$a_5 = -\frac{4 \cdot 3 + 2}{5 \cdot 4} a_3 = -\frac{7}{10} a_3 = \frac{7}{10}$$

$$\therefore y = 1 + x - x^2 - x^3 + \frac{5}{6}x^4 + \frac{7}{10}x^5$$

$$y(0.25) = 1.1758$$

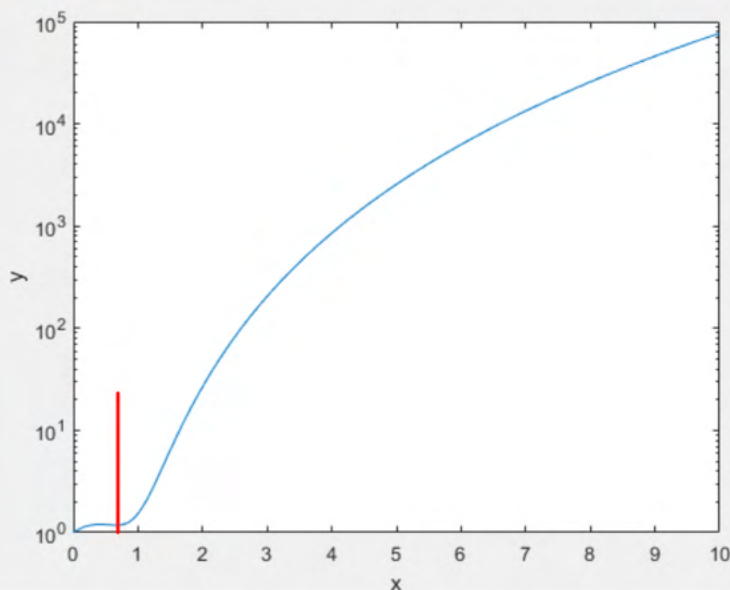
poly1.m

```
1 x = linspace(0,10,1000);
2 y = 1 + x - x.^2 - x.^3 + 5/6*x.^4 + 0.7*x.^5;
3 plot(x,y)
4 ylabel('log')
5 ylabel('y')
6 xlabel('x')
```

Figure 1

파일(F) 편집(E) 보기(V) 삽입(I) 툴(T) 데스크탑(D) 창(W) 도움말(H)

Figure 1



S.2

$$1. y_1 = 1 - \frac{1}{3}x^3 + \frac{1}{3}x^5 + \dots = \frac{1}{2} \ln \frac{1+x}{1-x}$$

$$y_2 = p_0(x) = 1$$

$$\text{let } y' = z$$

$$(1-x^2)y'' - 2xy' = 0$$

$$(1-x^2)\frac{dz}{dx} - 2xz = 0$$

$$\frac{dz}{z} = \frac{-2x}{1-x^2} dx$$

$$\ln|z| = -\ln|1-x^2| + C_0$$

$$z = \frac{C_1}{1-x^2} = y'$$

$$\frac{dy}{dx} = \frac{C_1}{1-x^2} = \frac{C_1}{2} \left(\frac{1}{1-x} + \frac{1}{1+x} \right)$$

$$\therefore y = \frac{C_1}{2} \ln \frac{1+x}{1-x} + C_2$$

$\therefore y_1, y_2$ 가 $\frac{1}{2}$ 을 basis이다.

$$2. y_1 = 1 - x^2 - \frac{1}{3}x^4 - \frac{1}{5}x^6 - \dots = 1 - \frac{1}{2}x \ln \frac{1+x}{1-x}$$

$$y_2 = p_1(x) = x$$

$$(1-x^2)y'' - 2xy' + 1.2y = 0$$

$$\text{let } y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1} = a_1 + 2a_2 x + 3a_3 x^2 + \dots$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = 2 \cdot 1 a_2 + 3 \cdot 2 a_3 x + 4 \cdot 3 a_4 x^2 + \dots$$

이제

$$(1-x^2) (2 \cdot 1 a_2 + 3 \cdot 2 a_3 x + 4 \cdot 3 a_4 x^2 + \dots)$$

$$- 2x (a_1 + 2a_2 x + 3a_3 x^2 + \dots)$$

$$+ 1.2 (a_0 + a_1 x + a_2 x^2 + \dots) = 0$$

$$x^0: 2 \cdot 1 a_2 + 2a_0 = 0$$

$$a_2 = -a_0$$

$$x^1: 3 \cdot 2 a_3 - 2a_1 + 2a_1 = 0$$

$$a_3 = 0$$

$$x^2: 4 \cdot 3 a_4 - 2 \cdot 1 a_2 - 4a_2 + 2a_2 = 0$$

$$a_4 = \frac{1}{3} a_2 = -\frac{1}{3} a_0$$

$$x^3: (5 \cdot 2)(5 \cdot 1) a_{5/2}$$

$$- 5(5 \cdot 1) a_5$$

$$- 2 \cdot 5 a_5$$

$$+ 2a_5 = 0$$

$$a_{5/2} = \frac{5-1}{5+1} a_5$$

$$a_3 = 0 \cdot a_1 = 0$$

$$a_4 = \frac{1}{3} a_2 = -\frac{1}{3} a_0$$

$\vdots$

$$\therefore y = a_0 + a_1 x + a_2 x^2 + \dots$$

$$= a_0 \left(1 - x^2 - \frac{1}{3}x^4 - \frac{1}{5}x^6 - \dots \right) + a_1 x$$

$\therefore y_1, y_2$ 가 $\frac{1}{2}$ 을 basis이다.

5.3

4. $xy' - y = 0$

$x^2 y'' - xy' = 0$

$b(x) = 0, c(x) = -1/x$

$b_0 = 0, c_0 = 0$

$\therefore h(h-1) = 0$

$h = 0, 1$

$\therefore y_1 = \sum_{n=0}^{\infty} a_n x^{n+1}$

$y_2 = y_1 \ln x + \sum_{n=0}^{\infty} A_n x^n = y_1 \ln x + \sum_{n=0}^{\infty} A_n x^n$

$\dot{y}_1 = \sum_{n=2}^{\infty} n(n+1) a_n x^{n-1}$

$\ddot{y}_1 = \sum_{n=2}^{\infty} n(n+1) a_n x^{n-2} = \sum_{n=0}^{\infty} a_{n+2} x^n$

$x^2: 5(5+1)a_5 - a_{5-1} = 0$

$a_5 = \frac{1}{5(5+1)} a_{5-1}$

$a_1 = \frac{1}{2} a_0 = \frac{1}{2} \quad \leftarrow c_0 = 1$

$a_2 = \frac{1}{2 \cdot 3} a_1 = \frac{1}{2 \cdot 2 \cdot 3} a_0$

$a_3 = \frac{1}{3 \cdot 4} a_2 = \dots$

$\vdots$

$\therefore y_1 = x + \frac{1}{2}x^2 + \frac{1}{12}x^3 + \dots$

$y_2 = y_1 \ln x + \sum_{n=0}^{\infty} A_n x^n$

$= y_1 \ln x + 1 - \frac{7}{4}x - \frac{7}{36}x^2 - \dots$

5. $x^2 y'' + x(2x-1)y' + (2x+1)y = 0$

$y'' + \frac{2x-1}{x}y' + \frac{x+1}{x^2}y = 0$

$b(x) = 2x-1, c(x) = 2x+1$

$b_0 = -1, c_0 = 1$

$\therefore h(h-1) - h + 1 = 0$

$h = 1 \quad \left(\frac{3}{2} \right)$

$\therefore y_1 = x \sum_{n=0}^{\infty} a_n x^n$

$= \sum_{n=0}^{\infty} a_n x^{n+1}$

$\dot{y}_1 = \sum_{n=1}^{\infty} (n+1) a_n x^n$

$\ddot{y}_1 = \sum_{n=2}^{\infty} (n+1)n a_n x^{n-1}$

clap

$x^2 \sum_{n=0}^{\infty} h(h+1) a_n x^{n-1} + x(2x-1) \sum_{n=0}^{\infty} (n+1) a_n x^n + (2x+1) \sum_{n=0}^{\infty} a_n x^{n+1} = 0$

$\therefore a_5 = -\frac{25+1}{5^2} a_{5-1}$

$a_1 = -3a_0 = -3$

$a_2 = -\frac{5}{4}a_1 = \frac{15}{4}$

$a_3 = -\frac{7}{9}a_2 = -\frac{35}{12}$

$\vdots$

$\therefore y_1 = x - x^2 + \frac{15}{4}x^3 - \frac{35}{12}x^4 + \dots$

$\therefore y_2 = y_1 \ln x + 4x^2 - \frac{29}{4}x^3 + \frac{29}{4}x^4 - \dots$

5.4

2. $x^2 y'' + xy' + (x^2 - \frac{1}{4})y = 0$
 $\nu^2 = \frac{1}{4}$
 $\nu = \frac{1}{2}, -\frac{1}{2}$
 $\therefore y = C_1 J_{1/2}(x) + C_2 J_{-1/2}(x)$

5.5

1. $x^2 y'' + xy' + (x^2 - 4)y = 0$
 $\nu^2 = 4$
 $\nu = -3, 3$
정칙이므로 basis는 $J_3(x), Y_3(x)$ 이다.
 $\therefore y = C_1 J_3(x) + C_2 Y_3(x)$

Review

2. Power-series Method는 수열에서 계를 Analytic이라고 하는
Frobenius Method는 Analytic이 아니므로 Series Solution을 구할 수
있을 것이다.

11. $y'' + 4y = 0$

let $y = \sum_{n=0}^{\infty} a_n x^n$
 $\ddot{y} = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$
대입하고 x^5 의 계수를 구한다.

$x^5: (s+2)(s+1)a_{s+2} + 4a_s = 0$
 $a_{s+2} = -\frac{4}{(s+2)(s+1)} a_s$
 $a_2 = -\frac{4}{2 \cdot 1} a_0 = -\frac{4}{2} a_0$
 $a_3 = -\frac{4}{3 \cdot 2} a_1 = -\frac{3}{2} a_1$
 $a_4 = -\frac{4}{4 \cdot 3} a_2 = -\frac{8}{8} a_0$
 $a_5 = -\frac{4}{5 \cdot 4} a_3 = -\frac{27}{40} a_1$

$\therefore y = a_0 \left(1 - \frac{4}{2} x^2 - \frac{8}{8} x^4 - \dots \right) + a_1 \left(x - \frac{3}{2} x^3 - \frac{27}{40} x^5 - \dots \right)$

20. $xy'' + y' - xy' = 0$

$y'' + \frac{1}{x}y' - \frac{x^2}{x^2}y' = 0$
 $b(x) = 1 \quad c(x) = -x^2$
 $\therefore b_0 = 1, c_0 = 0$
 $\therefore t(t-1) + t = 0$
 $t = 0 \quad (\frac{2}{3})$

$y = \sum_{n=0}^{\infty} a_n x^n$
 $\dot{y} = \sum_{n=1}^{\infty} n a_n x^{n-1}$
 $\ddot{y} = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$

대입하면

$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + \sum_{n=1}^{\infty} n a_n x^{n-1} - \sum_{n=0}^{\infty} a_n x^{n+1}$

x^0 의 계수: $a_1 = 0$

x^s 의 계수: $(s+1)s a_{s+1} + (s+1)a_{s+1} - a_{s-1} = 0$

$a_{s+1} = \frac{1}{(s+1)^2} a_{s-1}$

$a_{s+2} = \frac{1}{(s+2)^2} a_s$

...

$a_2 = \frac{1}{4} a_0$

$a_3 = \frac{1}{5^2} a_1 = 0$

$a_4 = \frac{1}{6^2} a_2$

$a_5 = \frac{1}{7^2} a_3 = 0$

...

$\therefore y_1 = 1 + \frac{1}{4}x^2 + \frac{1}{64}x^4 + \dots$

$\therefore y_2 = y_1 \ln x - 2x - \frac{5}{18}x^3 - \frac{85}{1260}x^5 - \dots$

6.1

1. $2t+8$

$$\mathcal{L}(2t+8) = 2\mathcal{L}(t) + 8\mathcal{L}(1)$$

$$= \frac{2}{s^2} + \frac{8}{s}$$

3. $\cos 2t$

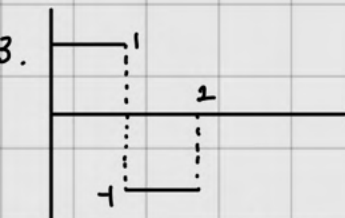
$$\mathcal{L}(\cos 2t) = \frac{s}{s^2 + (2)^2}$$

6. $e^{-t} \sinh 4t$

$$\mathcal{L}(e^{-t} \sinh 4t) = \mathcal{L}\left(e^{-t} \times \frac{e^{4t} - e^{-4t}}{2}\right)$$

$$= \frac{1}{2} \mathcal{L}(e^{3t} + e^{-5t})$$

$$= \frac{1}{2} \left(\frac{1}{s-3} + \frac{1}{s+5} \right)$$



$$f(x) = \begin{cases} 1 & (0 \leq x < 1) \\ -1 & (1 \leq x < 2) \end{cases}$$

$$\therefore \mathcal{L}(f(x)) = \int_0^1 e^{-st} dt + \int_1^2 -e^{-st} dt$$

$$= -\frac{1}{s} e^{-st} \Big|_0^1 + \frac{1}{s} e^{-st} \Big|_1^2$$

$$= -\frac{1}{s} e^{-s} + \frac{1}{s} + \frac{1}{s} e^{-2s} - \frac{1}{s} e^{-s}$$

$$= \frac{1}{s} (e^{-2s} - 2e^{-s} + 1) = \frac{1}{s} (e^{-s} - 1)^2$$

25. $\frac{0.294}{s^2 + 1.96} = \frac{0.25}{s^2 + 1.96} + \frac{1.4}{s^2 + 1.96}$

$$= 0.2 \mathcal{L}(\cos 1.4t) + \mathcal{L}(\sin 1.4t)$$

$$= \mathcal{L}(0.2 \cos 1.4t + \sin 1.4t)$$

$$\therefore f(t) = 0.2 \cos 1.4t + \sin 1.4t$$

31. $\frac{-s+11}{s^2-2s-3}$ $-(s-3)+8$ $s+1-12$

$$= \frac{-s+11}{(s-3)(s+1)} = \frac{1}{4} \left(\frac{-s+11}{s-3} - \frac{-s+11}{s+1} \right)$$

$$= \frac{1}{4} \left(-1 + \frac{8}{s-3} + 1 - \frac{12}{s+1} \right)$$

$$= \frac{1}{4} [\mathcal{L}(8e^{3t}) - \mathcal{L}(12e^{-t})]$$

$$\therefore f(t) = \frac{1}{4} (8e^{3t} - 12e^{-t})$$

$$= 2e^{3t} - 3e^{-t}$$

35. $2e^{-1/2t} \sin 4t$

$$\rightarrow 2 \sin 4at \approx \frac{1}{2} \text{ s-shift}$$

$$\therefore \mathcal{L}(2 \sin 4at) = \frac{2 \times 4a}{s^2 + (4a)^2}$$

$$\therefore \mathcal{L}(2e^{-\frac{1}{2}t} \sin 4at) = \frac{8a}{(s+\frac{1}{2})^2 + (4a)^2}$$

6.2

3. $y'' + y' - 6y = 0$ $y(0)=1, y'(0)=1$

L.T. $s^2 Y(s) - s y(0) - y'(0) + s Y(s) - y(0) - 6 Y(s) = 0$

$$(s^2 + s - 6)Y - s - 1 - 1 = 0$$

$$Y = \frac{s+2}{(s+3)(s-2)}$$

$$= \frac{1}{5} \left(\frac{s+2}{s-2} - \frac{s+2}{s+3} \right)$$

$$= \frac{1}{5} \left(\frac{4}{s-2} + \frac{1}{s+3} \right)$$

$$= \frac{4}{5} \mathcal{L}(e^{2t}) + \frac{1}{5} \mathcal{L}(e^{-3t})$$

$$\therefore y = \frac{4}{5} e^{2t} + \frac{1}{5} e^{-3t}$$

6. $y'' - 6y' + 5y = 29 \cos 2t$, $y(0)=3.2$, $y'(0)=2.2$

$$s^2 Y - s y(0) - y'(0) - 6s Y + 6 y(0) + 5 Y = 29 \times \frac{s}{s^2 + 2^2}$$

$$(s^2 - 6s + 5)Y - 3.2s - 2.2 + 19.2 = \frac{29s}{s^2 + 4}$$

$$(s-5)(s-1)Y = \frac{29s}{s^2 + 4} + 3.2s - 13$$

$$(s^2 + 4)(3.2s - 13) + 29s$$

$$Y = \frac{29s}{(s^2 + 4)(s - 6s + 5)} + \frac{3.2s - 13}{(s-5)(s-1)}$$

$$= \frac{1}{-6s+1} \left(\frac{29s}{s^2+4} - \frac{29s}{(s-5)(s-1)} \right) + \frac{1}{4} \left(\frac{3.2s-13}{s-5} - \frac{3.2s-13}{s-1} \right)$$

$$= \frac{29}{4(s^2+4)} - \frac{29}{4(s-5)(s-1)} + \frac{3.2s-13}{4(s-5)} - \frac{3.2s-13}{4(s-1)}$$

17. $f(t) = t e^{at}$, $f(0)=0$

$$f(t) = e^{at} + a t e^{at}$$

L.T.

$$\mathcal{L}(f(t)) = \mathcal{L}(t e^{at}) = \frac{1}{(s-a)^2}$$

$$\mathcal{L}(f) = sF(s) - f(0) = \frac{1}{s-a} + aF(s)$$

$$F(s) = \frac{1}{(s-a)^2}$$

same

6.3

4. Cos2t (0 < t < π)

$$\cos 2t (u(t) - u(t-\pi)) = \begin{cases} 0 & t < 0 \text{ or } \pi < t \\ \cos 2t & 0 < t < \pi \end{cases}$$

L.T

$$\downarrow$$
$$\frac{s}{s^2+2^2} - e^{-\pi s} \frac{s}{s^2+2^2}$$
$$= \frac{s}{s^2+2^2} (1 - e^{-\pi s})$$

14. $4(e^{-2s} - 2e^{-5s})/s$

$$= 4e^{-2s} \times \frac{1}{s} - 2e^{-5s} \times \frac{1}{s}$$

$$= 4\mathcal{L}(u(t-2)) - 2\mathcal{L}(u(t-5)) = \begin{cases} 0 & t < 2 \\ 4 & 2 < t < 5 \\ -4 & 5 < t \end{cases}$$

18. $4y'' - 12y' + 9y = 0$ $y(0) = \frac{2}{3}$, $y'(0) = 1$

$$4s^2Y - 4sy(0) - 4y'(0) - 12sY + 12y(0) + 9Y = 0$$

$$(4s^2 - 12s + 9)Y - \frac{8}{3}s - 4 - \frac{24}{3} = 0$$

$$Y = \frac{\frac{8}{3}s + 12}{4s^2 - 12s + 9} = \frac{4(2s+3)}{3(2s-3)^2} = \frac{4}{3} \times \frac{1}{2s-3}$$
$$= \frac{2}{3} \times \frac{1}{s-\frac{3}{2}}$$
$$= \frac{2}{3} \mathcal{L}(e^{\frac{3}{2}t})$$

$$\therefore y = \frac{2}{3} e^{\frac{3}{2}t}$$

14. $y'' - 6y' + 8y = e^{-t} - e^{-4t}$ $y(0) = 1$, $y'(0) = 4$

$$s^2Y - sy(0) - y'(0) - 6sY + 6y(0) + 8Y = \frac{1}{s+1} - \frac{1}{s+4}$$

$$(s^2 - 6s + 8)Y - 3 - 4 + 6 = \frac{1}{s+1} - \frac{1}{s+4}$$

$$(s-4)(s-2)Y - 5 + 2 =$$

$$Y = \frac{1}{(s+1)(s-4)(s-2)} - \frac{+1}{(s+4)(s-4)(s-2)} + \frac{1}{s-4}$$

$$= -\frac{1}{3} \left(\frac{1}{(s+1)(s-4)} - \frac{1}{(s-4)(s-2)} \right) + \frac{1}{6} \left(\frac{1}{(s+4)(s-4)} - \frac{1}{(s-4)(s-2)} \right) + \frac{1}{s-4}$$

$$= -\frac{1}{3} \left\{ -\frac{1}{5} \left(\frac{1}{s+1} - \frac{1}{s-4} \right) - \frac{1}{2} \left(\frac{1}{s-4} - \frac{1}{s-2} \right) \right\} + \frac{1}{6} \left\{ -\frac{1}{8} \left(\frac{1}{s+4} - \frac{1}{s-4} \right) - \frac{1}{2} \left(\frac{1}{s-4} - \frac{1}{s-2} \right) \right\} + \frac{1}{s-4}$$

$$= \frac{1}{15} \times \frac{1}{s+1} + 1.0375 \frac{1}{s-4} + \frac{1}{12} \times \frac{1}{s-2} - \frac{1}{48} \times \frac{1}{s+4}$$

$$\therefore y = \frac{1}{15} e^{-t} + 1.0375 e^{4t} + \frac{1}{12} e^{2t} - \frac{1}{48} e^{-4t}$$

24. $y'' + 3y' + 2y = \begin{cases} 1 & (0 < t < 1) \\ 0 & t > 1 \end{cases}$

L.T. $y(0) = 0$, $y'(0) = 0$

↓

$$s^2Y - sy(0) - y'(0) + 3sY - 3y(0) + 2Y = \frac{1}{s}$$

$$(s^2 + 3s + 2)Y = \frac{1}{s}$$

$$Y = \frac{1}{s(s+1)(s+2)}$$

$$= \frac{1}{2} \left(\frac{1}{s(s+1)} - \frac{1}{(s+1)(s+2)} \right)$$

$$= \frac{1}{2} \left\{ \frac{1}{s} - \frac{1}{s+1} - \left(\frac{1}{s+1} - \frac{1}{s+2} \right) \right\}$$

$$\therefore y = \frac{1}{2} - \frac{1}{2}e^{-t} - \frac{e^{-t}}{2} + \frac{e^{-2t}}{2}$$

$$= \frac{1}{2} - e^{-t} + \frac{e^{-2t}}{2} \quad (0 < t < 1)$$

L.T. $t > 1$

$$(s^2 + 3s + 2)Y = 0$$

$$Y = 0$$

$$y = 0$$

6.4

$$6. \quad y'' + 4y' + 5y = \delta(t-1), \quad y(0)=0, \quad y'(0)=3$$

$$s^2 Y - sy(0) - y'(0) + 4sY - 4y(0) + 5Y = e^{-s}$$

$$(s^2 + 4s + 5)Y - 3 = e^{-s}$$

$$Y = \frac{e^{-s}}{s^2 + 4s + 5} + \frac{3}{s^2 + 4s + 5}$$

$$= e^{-s} \times \frac{1}{(s+2)^2 + 1} + \frac{3}{(s+2)^2 + 1}$$

↓ inverse L.T.

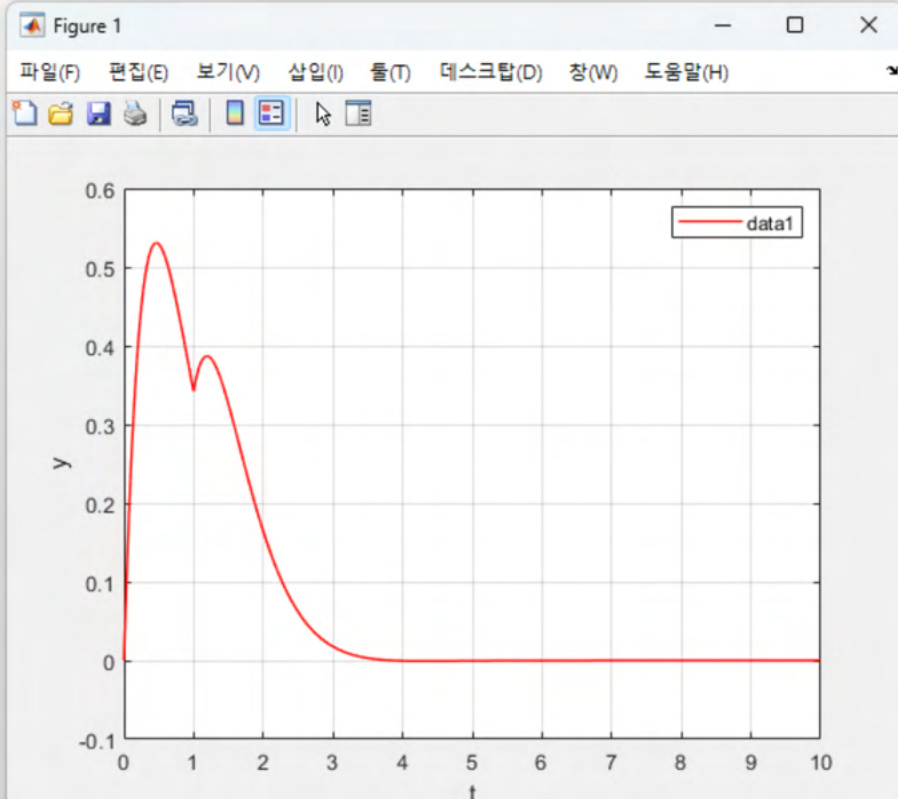
$$y(t) = e^{-2(t-1)} \sin(t-1) u(t-1) + 3e^{-2t} \sin t$$

$$= \begin{cases} 3e^{-2t} \sin t & (0 < t < 1) \\ e^{-2(t-1)} \sin(t-1) u(t-1) + 3e^{-2t} \sin t & (1 \leq t) \end{cases}$$

code5_6.m

```
t = linspace(0,10,1000);
y = zeros(size(t));
n = numel(t);
for i = 1 : n-1
    if t(i) < 1
        y(i) = 3 * exp(-2*t(i)) * sin(t(i));
    elseif t(i) >= 1
        y(i) = exp(-2*(t(i)-1)) * sin(t(i)-1) + 3 * exp(-2*t(i)) * sin(t(i));
    end
end

plot(t,y,'-r',linewidth = 1)
legend()
xlabel("t")
ylabel("y")
grid on
```



6.5

2. $\int_0^t \sin w(t-\tau) d\tau$

$$= \int_0^t \sin w(t-\tau) d\tau$$

$$= \left. -\frac{1}{w} \cos w(t-\tau) \right|_0^t$$

$$= -\frac{1}{w} \cos wt + \frac{1}{w} \cos w(t-0)$$

8. $y(t) + 4 \int_0^t y(\tau) (t-\tau) d\tau = 2t$

L.T. $\downarrow$ $y(t) + 4 \int_0^t y(\tau) (t-\tau) d\tau = 2t$

$$Y(s) + 4Y(s) \times \frac{1}{s^2} = \frac{2}{s^2}$$

$$Y(s) = \frac{2}{s^2 + 4}$$

$$\mathcal{L}^{-1}(Y(s)) = \mathcal{L}^{-1}\left(\frac{2}{s^2 + 2^2}\right)$$

$$\underline{y = \sin 2t}$$

17. $\frac{5.5}{(s+1.5)(s-4)}$

$$\Rightarrow \mathcal{L}^{-1}\left(\frac{5.5}{(s+1.5)(s-4)}\right) = 5.5 e^{-1.5t} \times e^{4t}$$

$$= 5.5 \int_0^t e^{-1.5\tau} \times e^{4(t-\tau)} d\tau$$

$$= 5.5 \int_0^t e^{-5.5\tau} \times e^{4t} d\tau$$

$$= -e^{4t} \times e^{-5.5\tau} \Big|_0^t$$

$$= -e^{1.5t} + e^{4t}$$

6.6, 6.7 48%

Review

$$12. e^{-2t} (\cos 2t - 4 \sin 2t)$$

$$= e^{-2t} \cos 2t - 4 e^{-2t} \sin 2t$$

↓ L.T.

$$= \frac{s+2}{(s+2)^2 + 2^2} - 4 \times \frac{2}{(s+2)^2 + 2^2}$$

$$= \frac{s-6}{(s+2)^2 + 2^2}$$

$$16. 4(t-2\pi) \cos 2t$$

↓ L.T.

$$= \frac{s}{s^2 + 4^2} e^{-2\pi s}$$

$$27. \frac{2s+1}{s^2 + 2s + 5}$$

$$= 2 \times \frac{s+1 - \frac{1}{2}}{(s+1)^2 + 4}$$

$$= 2 \times \frac{s+1}{(s+1)^2 + 4} - \frac{2}{(s+1)^2 + 4} \times \frac{1}{2}$$

↓ Inverse L.T.

$$= 2 \times e^{-t} \cos 2t - \frac{1}{2} e^{-t} \sin 2t$$

30. 48%